

that spread through the air, triggering a response from animals some distance away.

PHOTOSYNTHESIS

A process whereby organisms use light energy to make food and oxygen; photosynthesis occurs in plants, algae, and many micro-organisms.

PLACENTA

An organ developed by an embryo mammal that allows it to absorb nutrients and oxygen from its mother's bloodstream before it is born.

PLANKTON

Floating organisms, many of them microscopic, that drift in open waters, particularly near the surface of the sea. Planktonic organisms can often move, but most are too small to make any headway against strong currents. Planktonic animals are known as zooplankton. Planktonic algae are called phytoplankton.

PLASTRON

The lower part of the shell structure of tortoises and turtles.

POLLEN

Tiny grains produced by seed plants that contain male gametes for fertilizing the female egg of either a flowering plant or a coniferous plant.

POLYGAMOUS

A reproductive system in which individuals mate with more than one partner during the course of a single breeding season.

PREDATOR

An animal that catches and kills other animals, referred to as its prey. Some predators attack and catch their prey by lying in wait, but most actively pursue and attack other animals.

PREHENSILE

Able to curl around objects and grip them.

PREMOLAR (TOOTH)

In mammals, a tooth positioned midway along the jaw between the canine and the molar teeth. In carnivores, specialized premolar teeth act like shears, slicing through flesh.

PROBOSCIS

An animal's nose, or set of mouthparts with a nose-like shape. In insects that feed on fluids, the proboscis is often long and slender, and can usually be stowed away when not in use.

PROKARYOTE

An organism whose cells do not have a nucleus. The Archaea and Bacteria used to be classified as Prokaryota.

PRONOTUM

The part of an insect's cuticle that covers the first segment of its thorax, often hardening into a shell.

PROTEIN

Substance found in food, such as meat, fish, cheese, and beans, that is used for growth and for carrying out a variety of essential biological functions.

PSEUDOPOD

Temporary projection of a cell, such as an amoeba or a white blood cell, which is used for creeping forward or for catching food.

PUPA (PL. PUPAE)

In insects, a stage during which the larval body is broken down and rebuilt as an adult. During the pupal stage the insect does not feed and usually cannot move, although some pupae may wriggle if they are touched. The pupa is protected by a hard case, which itself is sometimes wrapped in silk.

QUADRUPEDAL

An animal that walks on four legs.

RAPTOR

A bird of prey.

RHIZOME

A creeping or underground stem that can send out new shoots.

ROCK

Material made up of one or more minerals.

ROOT NODULE

Spherical swelling on the root of a legume that contains nitrogen-fixing bacteria.

RUMINANT

A hoofed mammal that has a specialized digestive system with several stomach chambers. One of these – the rumen – contains large numbers of micro-organisms which help break down plant food. To speed this process, a ruminant usually regurgitates its food and rechews it, a process called chewing the cud.

SALLY

A bird's short flight from a perch to catch an invertebrate, often in mid-air.

SCUTE

A shield-like plate or scale that forms a bony covering on some animals.

SEDIMENTARY ROCK

Rock formed by the consolidation and hardening of rock fragments, organic remains, or other material.

SEED

Developmental stage of a seed plant, consisting of an encapsulated embryo.

SEPAL

One of the parts of the calyx of a flower, usually small and leaf-like and enveloping the unopened flower bud.

SEXUAL DIMORPHISM

Showing physical differences between males and females. In animals that have separate sexes, males and females always differ, but in highly dimorphic species, such as elephant seals, the two sexes look very different and are often unequal in size.

SHOOT

The aerial part of plant. It is a new growth, usually growing upwards.

SHRUB

Woody perennial plant with multiple stems.

SIMPLE LEAF

A leaf with an undivided blade.

SPERMATOPHORE

A packet of sperm that is transferred either directly from male to female, or indirectly – for example, by being left on the ground. Spermatophores are produced by a range of animals, including salamanders, squid, and some arthropods.

SPIRACLE

In some fish, an opening behind the eye that lets water flow into the gills. In insects and myriapods, the spiracle is an opening on the body wall that lets air into the tracheal system.

SPORE

A single cell containing half the quantity of genetic material of typical body cells. Unlike gametes, spores can divide and grow without being fertilized. Spores are produced by fungi, algae, and plants.

SPOROPHYTE

The spore-producing stage of a plant. It is the dominant (visible) stage of ferns and seed plants.

STAMEN

Male reproductive part of a flower. It has an anther borne on a long filament.

STEREOSCOPIC (VISION)

Vision in which the two eyes face forward, giving overlapping fields of view and allowing the animal to judge distance. Also called binocular vision.

STOMA (PL. STOMATA)

Tiny adjustable pore on the surface of a plant that allows the exchange of gases for photosynthesis and respiration.

SWIM BLADDER

A gas-filled bladder that most bony fish use to regulate their buoyancy. By adjusting the gas pressure inside the bladder, a fish can become neutrally buoyant, meaning that it neither rises nor sinks.

SYMBIOSIS

Any relationship between two different species within an ecological community. Examples of symbiotic relationships are predator–prey, parasite–host, and mutualism.

TEPAL

The outer part of a flower undifferentiated into sepals and petals. Tepals collectively make up the perianth.

TERRESTRIAL

Living wholly or mainly on the ground.

TERRITORY

An area defended by an animal, or group of animals, against other members of the same species. Territories often include resources, such as food supply, that help the male attract a mate.

THORAX

The middle region of an arthropod's body. The thorax contains powerful muscles and, if the animal has any, bears legs and wings.

In vertebrates with four limbs, the thorax is the chest.

TORPOR

A sleep-like state in which body processes slow to a fraction of their normal rate. Animals usually become torpid to survive difficult conditions, such as extreme cold or lack of food.

TRACHEAL SYSTEM

A system of minute tubes that arthropods (for example, insects) use to carry oxygen into their bodies. Air enters the tubes through openings called spiracles, and then flows through the tracheae to reach individual cells.

TREE

A woody perennial plant usually with a well-defined stem, or trunk, and a crown of branches above.

UNGULATE

A hoofed mammal.

UTERUS

In female mammals, the part of the body that contains and usually nourishes developing young. In placental mammals, the young are connected to the wall of the uterus via a placenta.

VECTOR

An organism that transmits a disease-causing parasite from one host to another.

VEIL

Thin skin- or web-like tissue that protects the fruitbody of a fungus.

VESTIGIAL

Relating to an organ that is atrophied or non-functional.

VIVIPAROUS

Reproducing by giving birth to live young.

VOLVA

A sac-like remnant of a veil at the base of a fungus's fruitbody stem.

WOODY PLANT

A plant that has wood – a type of strengthening tissue found in plants consisting of thick-walled water-transporting vessels.

ZYGODACTYL FEET

A specialized arrangement of the feet in which the toes are arranged in pairs, with the second and third toes facing forwards and the first and fourth toes facing backwards. This adaptation helps birds to climb and perch on tree-trunks and other vertical surfaces. Several groups of birds have zygodactyl feet, including parrots, cuckoos, turacos, owls, and toucans, woodpeckers, and their relatives.